

SIMATS ENGINEERING

(SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Scenario-Based

Course Code: ITA05

Course Title: Computer Vision

CO Assessed: CO2 – Application (BTL 3)

Assessment: Assessment Tool 2 – Scenario-Based

Weightage: 20% of CIA

Total Marks: 100

CO2 Statement: *Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)*

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Batch Date: 09-08-2026

Code of Conduct

I, Rajashree S (192521079), certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Rajashree S

RUBRICS FOR CALCULATION: ASSESSMENT TOOL 2 – Scenario-Based

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario	
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues	
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts	
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided	
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand	

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ASSESSMENT TOOL 2: Scenario-Based

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA05

COURSE OUTCOME COVERED

CO2: *Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)*

Assessment Tool Weightage: 20%

SCENARIO-BASED

4. Blurred Edges

An image processed for noise removal appears overly smooth, causing edges to blur.

- (a) Interpret the scenario and explain why edges are blurred.
- (b) Identify key issues related to smoothing and edge loss.
- (c) Apply an appropriate technique to restore edges.
- (d) Justify your decision with clear reasoning.
- (e) Structure your explanation clearly.

ANSWER

a. Scenario and Interpretation

When an image is processed to remove noise, smoothing filters (such as mean or Gaussian filters) average each pixel with its neighbors. Since noise and true edges both contain high-frequency intensity changes, the filter cannot separate them, so it smooths across edges as well. This spreads out the sharp intensity transition at object boundaries, making the edges appear blurred.

b. Key Issues

- Loss of high-frequency detail along with noise, since smoothing acts as a low-pass filter.
- Larger smoothing kernels remove more noise but blur edges more severely.
- Smoothing filters cannot distinguish between random noise and meaningful edge information.
- Blurred edges reduce accuracy of later steps such as edge detection and segmentation.
- Therefore, noise removal and edge sharpness must be balanced.

c. Technique Applied

The appropriate sequence is:

Noisy Image → Bilateral / Median Filtering → Edge-Preserved Smooth Image

- Apply Bilateral Filtering (or Median Filtering for impulse-type noise) instead of a plain mean/Gaussian filter.
- Bilateral filtering smooths using both spatial closeness and intensity similarity, so it avoids averaging across edges.
- If the image is already blurred, apply unsharp masking or Laplacian sharpening afterward to restore edge contrast.

d. Justification

Bilateral filtering is preferred over standard smoothing because it assigns low weight to neighboring pixels that differ greatly in intensity, which stops it from blending pixels across a true edge. This preserves boundary sharpness while still averaging out

noise within flat regions. Median filtering is justified as an alternative when the noise is impulse-type (salt-and-pepper), since it removes outlier values without significantly softening edges. Both methods directly address the root cause — the inability of ordinary smoothing filters to separate noise from edge information — rather than blurring first and attempting to recover detail afterward.

e. Final Interpretation

